

Smart Lost & Found System

Saleha Tariq

Project overview



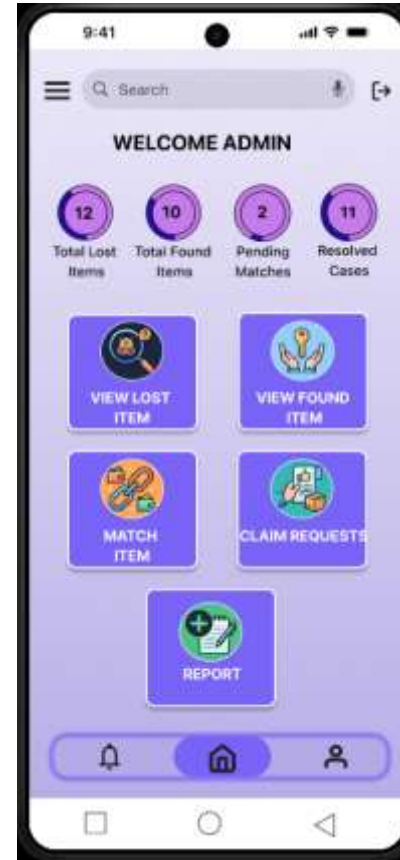
The product:

The Smart Lost & Found System is a web-based application that helps users report lost and found items and recover them efficiently. It connects users with administrators through a centralized platform that manages item tracking, matching, and claim verification.



Project duration:

27 April 2026 – 18 May 2026



Project overview



The problem:

Many people lose personal belongings in public places like universities, offices, and transport areas. There is no centralized system to report or find these items, which leads to loss, confusion, and wasted time.



The goal:

To design a system that allows users to easily report lost/found items, search for items, and recover them through a secure and verified claim process managed by an admin.

Project overview



My role:

UX Designer / UX Research Contributor



Responsibilities:

- Designed high-fidelity UI screens.
- Conducted user research activities.
- Assisted in usability testing sessions.
- Improved UI based on user feedback and testing results.

Understanding the user

- User research
- Personas
- Problem statements
- User journey maps

User research: summary



User research was conducted through informal interviews and observations. Initially, it was assumed that users mainly struggle with reporting lost items. However, research revealed that users also face difficulty in tracking item status and verifying ownership. This led to adding features like notifications and claim verification.

User research: pain points

1

Time-Consuming Manual Search

People often spend a lot of time physically searching different locations or asking others about lost belongings.

2

Fake Claim Requests

Some users may falsely claim ownership of found items, creating trust and security issues.

3

Difficulty Finding Matching Items

Users struggle to identify whether a found item matches their lost item because of incomplete information.

4

Delayed Verification Process

Manual verification by administrators can delay the return of items to their rightful owners.

Persona: Hania Ahmed

Problem statement:

Hania is a busy university student who needs a simple and secure way to report and recover lost items because traditional lost-and-found methods are slow, unorganized, and unreliable.



Hania

Age: 21 Years

Education: BA in English

Family: Parents and two younger siblings

Hometown: Karachi, Pakistan

Occupation: University Student

Persona: Hania Ahmed

"I just want an easy way to report my lost items and quickly know if someone has found them."

Goals:

- Quickly report lost items
- Find matching Items easily
- Receive updates about claim status
- Recover belongings securely

Frustrations:

- No proper platform for reporting lost Items
- Manual searching takes too much time
- Lack of notifications and tracking
- Fear of fake claims or lost records

Hania is a university student who frequently carries important items such as a phone, wallet, and ID card on campus. She wants a fast and reliable system where she can report lost items, search available found items, and receive updates without manually visiting different departments.

User journey map

[Hania wants a fast and reliable way to recover her lost belongings without wasting time searching manually. She expects the system to be simple, secure, and easy to navigate. During the journey, her emotions change from stress and confusion to relief after successful verification and item recovery. The design focuses on improving trust, reducing delays, and making communication between users and admins smoother.]

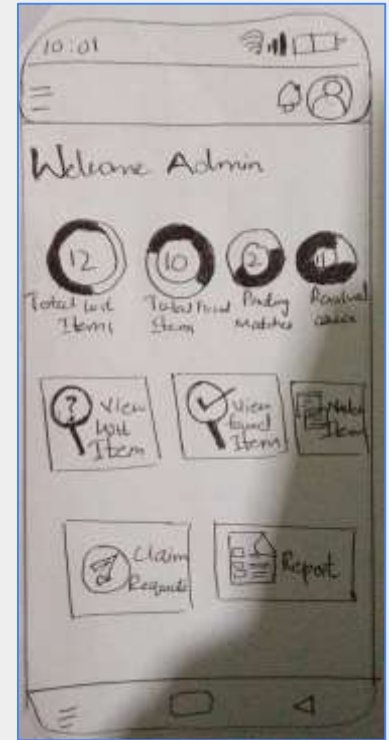
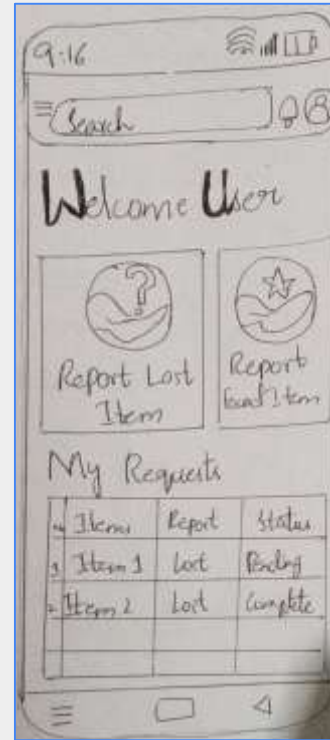
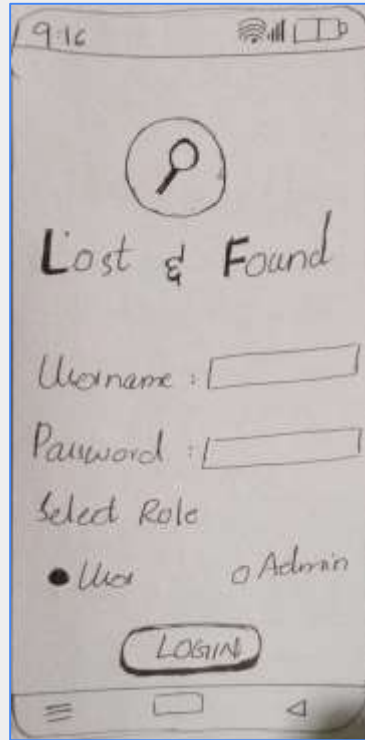
Persona: Hania Ahmed

Goal: Recover her lost item quickly and securely through an easy-to-use system.

ACTION	Report Lost Item	Search for Matching Item	Submit Claim Request	Verify Ownership	Receive Item
Task list	A. Open website/app B. Fill lost item form C. Upload item details/image	A. Use search filters B. Check matching items C. Compare descriptions	A. Open claim form B. Enter ownership details C. Submit request	A. Wait for admin response B. Provide proof if needed C. Track claim status	A. Receive approval notification B. Meet admin/user C. Collect item
Feeling adjective	- Worried - Confused	- Hopeful - Curious	- Nervous - Optimistic	- Impatient - Anxious	- Relieved - Happy
Improvement opportunities	- Simpler reporting form - Add image upload guide	- Better search filters - AI-based matching suggestions	- Clearer claim instructions - Faster submission process	- Real-time status updates - Faster admin verification	- Secure handover process - Feedback/rating system

Paper wireframes

Paper wireframes were created to quickly explore different layout ideas for the application. The focus was on simple navigation, easy item reporting, and quick access to search functionality. Multiple layout variations were tested before selecting the most user-friendly design.



9:31

Report Lost Item

Item Name:

Category:

Description:

Location:

Date:

9:32

Report Found Item

Item Name:

Description:

Location:

Image:

Date:

9:34

Search Items

Category: Location:

Search Results:

Item 1

Item 2

Item 3

9:45

Claim Item

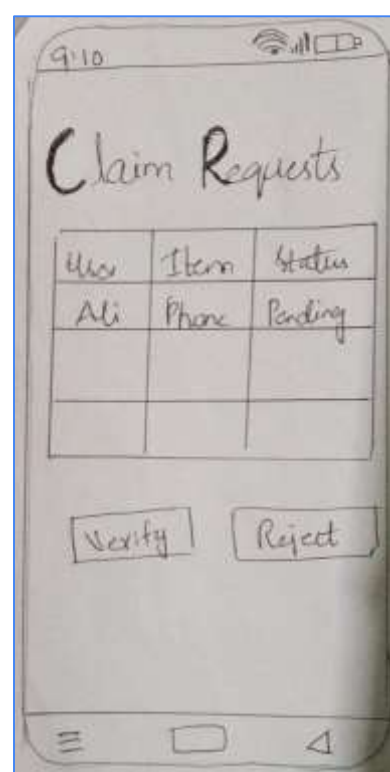
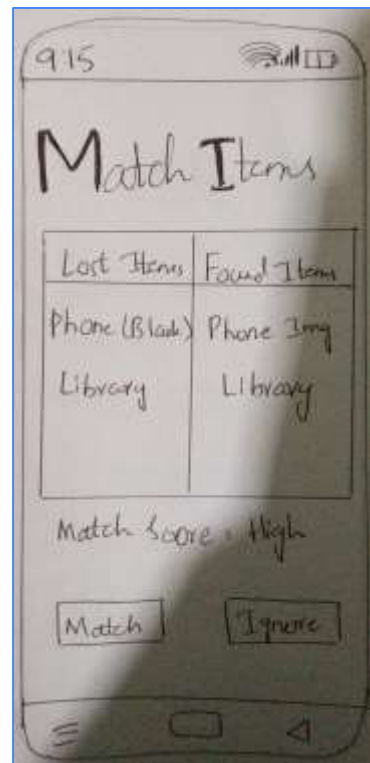
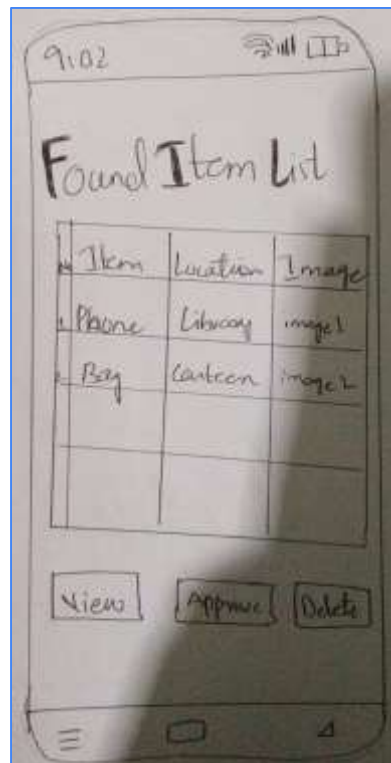
Item Name:

Category:

Location:

Status:

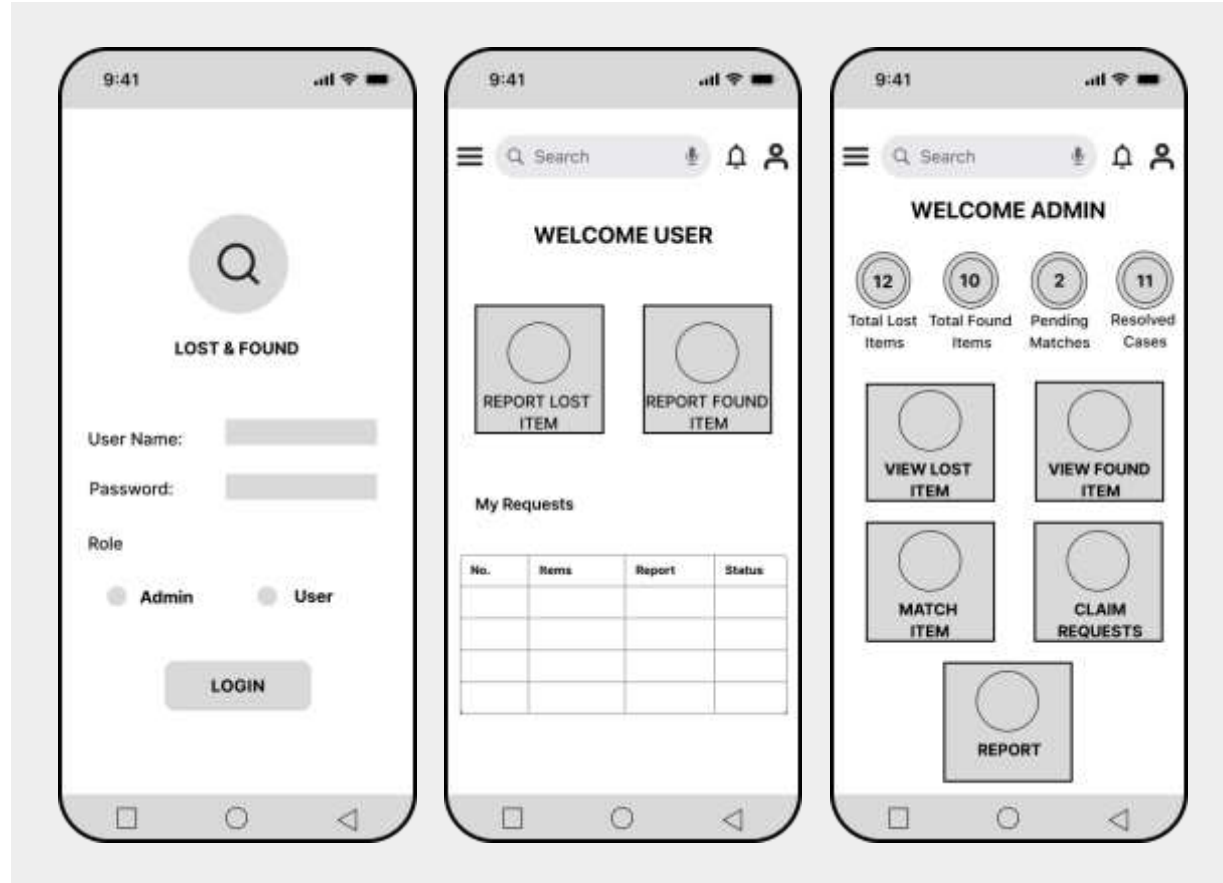
Description:



Low-fidelity prototype

[Link:

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9:41

LOST & FOUND

REPORT FOUND ITEM

Item Name:

Description:

Location:

Image:

Date:

SUBMIT

9:41

LOST & FOUND

REPORT LOST ITEM

Item Name:

Category:

Description:

Location:

Date:

SUBMIT

9:41

LOST & FOUND

Search

Category Filter

Search Results

Item 1

Item 2

Item 3

Item 4

9:41

LOST & FOUND

CLAIM ITEM

Item Name:

Description:

Location:

Category:

Status:

REQUEST CLAIM

9:41

LOST & FOUND

LOST ITEMS LIST

Lost Items

No.	Items	Location	Status

View Approve Delete

9:41

LOST & FOUND

FOUND ITEMS LIST

Found Items

No.	Items	Location	Image

View Approve Delete

9:41

LOST & FOUND

MATCH ITEMS

Lost Item	Found Item

Match Score:

Match Ignore

9:41

LOST & FOUND

CLAIM REQUESTS

No.	User	Item	Status

Verify Reject

Usability study: findings

Two rounds of usability studies were conducted with regular users to evaluate the ease of reporting items, searching for lost belongings, and understanding the claim process. Feedback from participants helped improve navigation, simplify forms, and make important features more visible.

Round 1 findings

- 1 Logout button was hard to find.
- 2 No navigation bar for easy movement.
- 3 Some users were confused about the item claim verification process.

Round 2 findings

- 1 Navigation bar was added.
- 2 Logout button placement was improved.
- 3 Interface became more user-friendly.

Refining the design

- Mockups
- High-fidelity prototype
- Accessibility

Mockups

Before Usability Study

- No navigation bar for switching pages.
- Logout option not easily visible.
- Overall layout looked cluttered.

After Usability Study

- Navigation bar added for easy page switching
- Logout button made visible and accessible
- Cleaner and more structured dashboard layout.

Before usability study



After usability study



Mockups

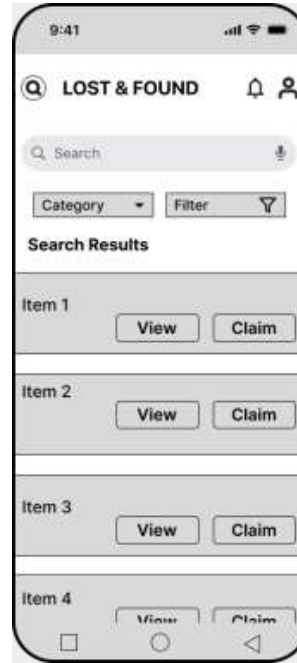
Before Usability Study

- Form fields were overcrowded and not properly structured.
- Only item name was visible, making it hard to understand full details.
- Poor spacing caused layout to look cluttered and confusing.

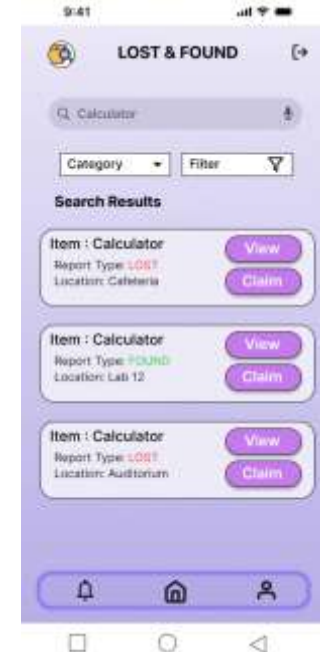
After Usability Study

- Form layout improved with better spacing and alignment.
- Key details like item type and location are now visible along with item name.
- Improved readability and clearer presentation of reported items.

Before usability study



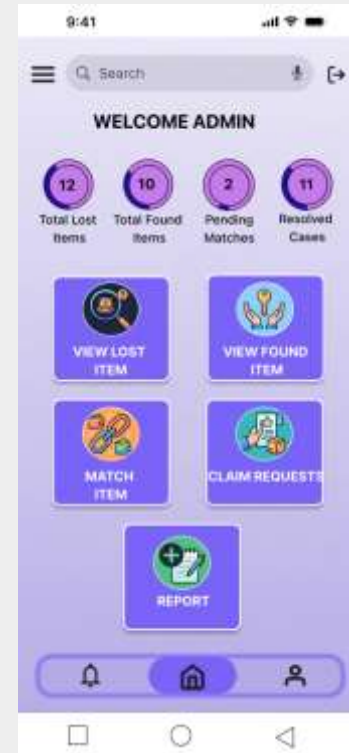
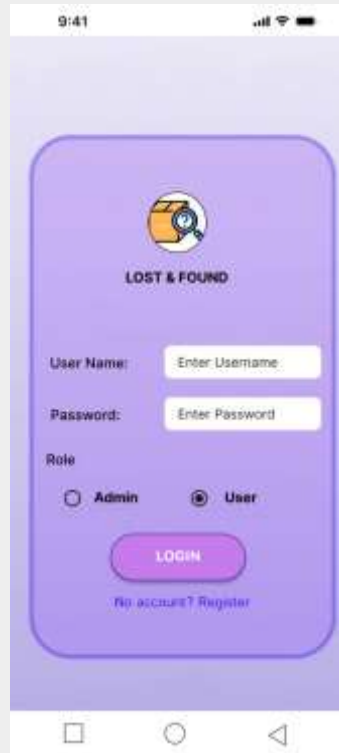
After usability study

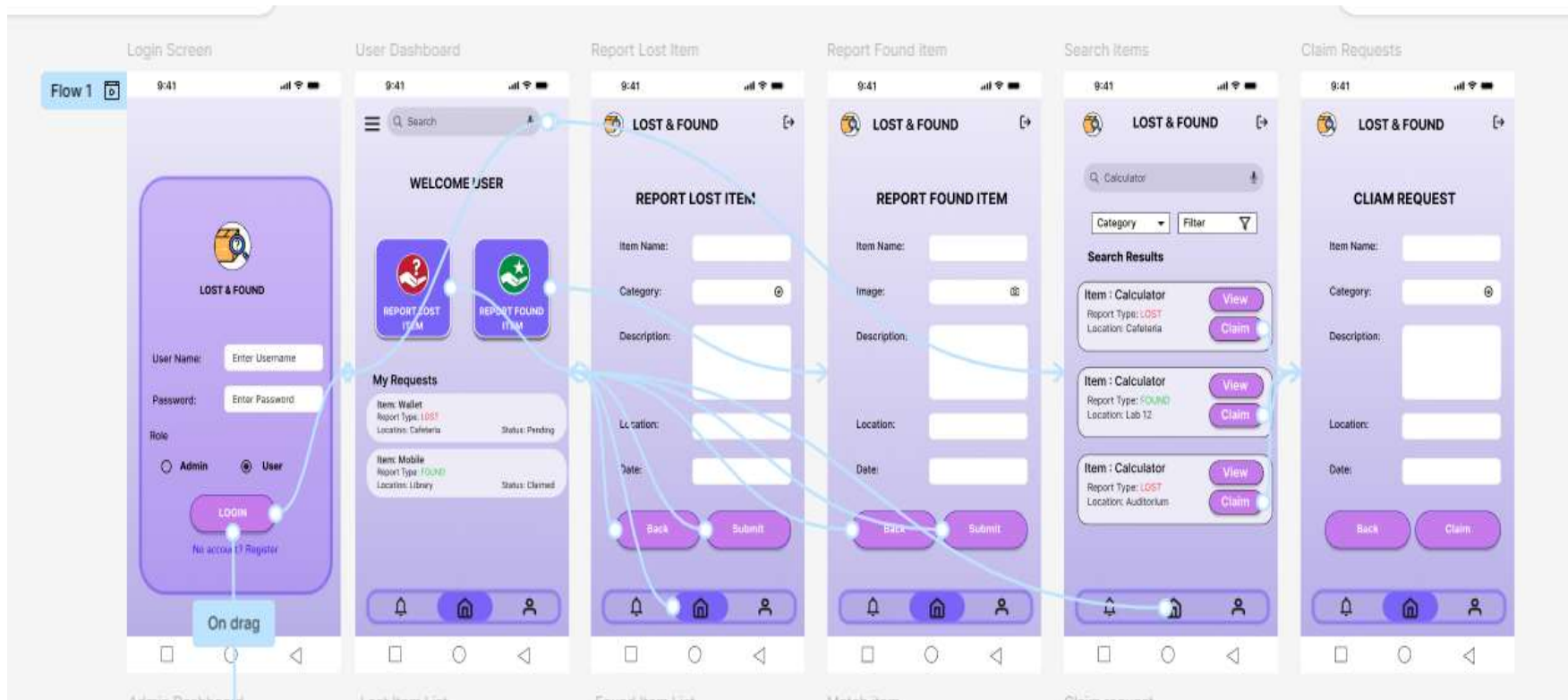


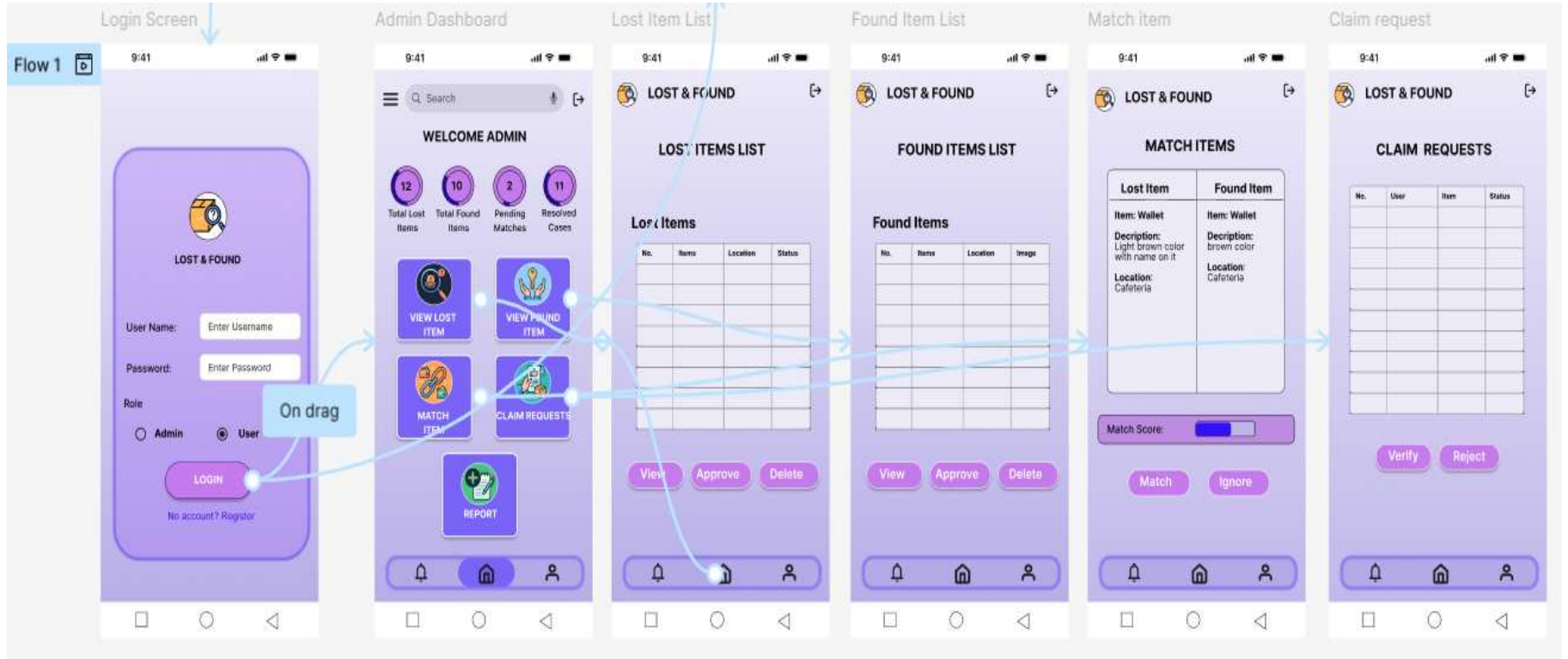
High-fidelity prototype

[Link:

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Accessibility considerations

1

High-contrast colors and readable fonts were used to improve visibility for users with visual impairments.

2

Buttons and navigation elements were designed with clear labels and sufficient spacing for easier interaction.

3

Simple language and organized layouts were used to make the system understandable for users with different technical skill levels.

Going forward

- Takeaways
- Next steps

Takeaways



Impact:

The Smart Lost & Found System provides users with a faster and more organized way to report and recover lost items. Feedback from users showed that the platform made the process easier, reduced confusion, and improved trust through secure claim verification.



What I learned:

Throughout this project, I learned the importance of user-centered design and continuous testing. Conducting usability studies helped me understand real user problems and improve the interface based on practical feedback. This also improved my skills in wireframing, prototyping, and designing accessible interfaces.

Next steps

1

Add AI-based item matching to automatically suggest similar lost and found items.

2

Implement email and SMS notifications to keep users updated about item status and verification.

3

Introduce a feedback and rating system for verified claims to improve trust and transparency.

Let's connect!



Thank you for reviewing our project portfolio. This project was developed as part of a team effort, and I am grateful for the opportunity to collaborate, learn, and contribute to building a user-focused solution.

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